Predicting Coronary Heart Disease from BRFSS with Pure-NumPy Ensembles

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***Abstract*—Predicting coronary heart disease (`_MICHD`) from 321 BRFSS survey features is challenging due to extreme class imbalance (8.8% positive) and non-random missingness. We develop an end-to-end classification pipeline strictly from scratch in pure NumPy. We resolve survey skip patterns via domain-guided imputation, benchmark the six course linear models against custom histogram-based gradient boosted trees (GBM) and neural networks (MLP), and optimize decision thresholds. Our final GBM-MLP ensemble achieves an F1 score of 0.447 (87.7% accuracy) on AIcrowd, substantially outperforming linear baselines. Crucially, we uncover severe target leakage in seven post-event survey questions: while exploiting them inflates test F1 to 0.457 (88.3% accuracy), we strictly exclude them from our primary model to ensure clinically sound pre-event screening.**

I. Introduction

Coronary heart disease is a leading global cause of death, motivating low-cost screening tools. The Behavioral Risk Factor Surveillance System (BRFSS) is an annual telephone survey assessing health conditions across the United States [1]. The binary target `_MICHD` indicates whether a respondent has experienced coronary heart disease or a heart attack. The dataset contains 328,135 training rows, 109,379 test rows, and 321 features. Predicting this outcome poses key challenges: extreme class imbalance (8.83% positive), non-standard encodings (refusal and frequency codes), and widespread missingness. A majority-class baseline yields 91.2% accuracy but zero F1; hence, all models are evaluated on positive-class F1 [2]. Constrained to pure NumPy implementations, our main contributions are: (i) codebook-informed preprocessing that resolves survey skip logic and missingness signals, (ii) a systematic benchmark of linear models, histogram-based gradient boosted trees (GBM), and neural networks (MLP) with calibrated decision thresholding, and (iii) the clinical discovery and audit of target leakage in post-event survey questions [3].

II. Data and preprocessing

Value and unit cleaning. Following the CDC codebook [4], sentinel value 88 ("none") is mapped to 0, while 7 and 9 ("don't know"/"refused") are treated as missing. Imperial and metric measurements (e.g., height, weight) are standardized to centimeters and kilograms.

Skip patterns and non-response. Over 45% of questions are $> 50\%$ missing due to survey skip logic rather than random omissions. For example, blood pressure medication (`BPMEDS`) is asked only of hypertensive respondents (16.3% prevalence vs. 3.8% when skipped). We apply eleven domain rules to infer implied negatives (e.g., no hypertension implies no medication). Because respondents answering "don't know" exhibit higher disease rates, we extract 61 binary uncertainty masks as predictive risk features.

Feature selection and pipeline. We remove 6 constant columns, administrative IDs, columns $> 95\%$ missing, and 44 collinear pairs ($|r| > 0.9$), retaining 75 raw features, 2 engineered metrics, and 61 uncertainty masks (138 total). Heavily skewed continuous variables are compressed via $\log(1 + x)$. We then perform median/mode imputation, standardization, one-hot encoding, and add 10 clinical interaction terms. To prevent data leakage, all preprocessing statistics are estimated strictly inside training folds.

III. Models and validation

Validation protocol. A stratified 20% split (65,627 rows) is held out for final evaluation; model selection and hyperparameter tuning use 5-fold stratified cross-validation (CV) on the remaining 262,508 development rows.

Linear baselines. We evaluate the six course baseline methods in `implementations.py` [5] on standardized features, using targets in ± 1 (mean squared error) and $\{0, 1\}$ (logistic loss).

Gradient boosted trees. Decision trees naturally capture non-linear interactions without requiring monotonic scaling. We implement a histogram-based gradient boosted tree (GBM) [6] in NumPy, minimizing logistic loss with L2-regularized Newton leaf steps ($\lambda = 5$) [7]. Features are grouped into 64 quantile bins with missing entries routed automatically. We use depth-5 trees, learning rate 0.05, subsampling (0.7 rows, 0.8 columns), target-encoded state codes [8], and early stopping.

Neural network. An MLP with 64 ReLU hidden units is implemented in NumPy to provide smooth continuous decision boundaries. Trained with Adam [9] (learning rate 10^{-3}, batch size 512, weight decay 10^{-3}), logits are averaged across 10 random seeds to eliminate initialization variance.

Ensemble and decision rule. We blend the tree ensemble and neural network by averaging raw output scores ($0.5\, s_{\text{GBM}} + 0.5\, s_{\text{MLP}}$). Rather than a fixed 0.5 probability cutoff, we classify the top 14% highest-risk individuals as positive. This percentile rule is invariant to calibration drift across folds; as shown in Fig. 1, F1 remains stable across positive rates of 12–16%.

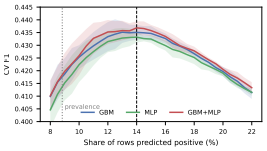


Figure 1. CV F1 (mean ± std over 5 folds) vs. predicted positive share. Dotted: prevalence (8.83%). Dashed: chosen cutoff (14%).

Table I
POSITIVE-CLASS F1 ACROSS MILESTONES (5-FOLD CV VS. 20% HELD-OUT VALIDATION SET).

Model & Configuration	CV F1	Held-out
Always -1 (majority class)	0.0000	0.0000
Logistic regression (0.5 cutoff)	0.2302 ± 0.0047	–
Course Baselines (Top 14% tuned cutoff):		
Least squares / Ridge / GD	0.4216 ± 0.0034	–
Linear regression (SGD, unconverged)	0.3853 ± 0.0085	–
Logistic regression (GD)	0.4302 ± 0.0047	–
Regularized logistic regression (GD)	0.4300 ± 0.0044	0.4277
Delivered NumPy Models (Top 14%):		
Histogram GBM (3 seeds)	0.4354 ± 0.0032	0.4343
Neural network MLP (10 seeds)	0.4334 ± 0.0026	0.4310
GBM + MLP Ensemble	$\mathbf{0.4368 \pm 0.0026}$	**0.4340**

IV. RESULTS

Table I benchmarks all models on standardized development folds:

Impact of decision thresholding. Due to class imbalance, the default 0.5 cutoff yields an F1 of only 0.2302. Selecting the top 14% highest-risk individuals immediately elevates F1 to 0.4302 (+0.200 gain), marking the single largest performance jump in the pipeline.

Impact of data preprocessing. Raw survey responses introduce severe numerical distortions: for instance, code 88 (denoting zero sick days) is treated as a high-risk value of 88 if uncleaned. Resolving codebook sentinels (88 to 0, 7/9 to missing), reconstructing eleven skip-logic branches, and log-compressing activity outliers raise the tuned logistic regression CV F1 from 0.4277 to 0.4311, establishing a robust baseline for modeling.

Model comparison and ensembling. Least squares, ridge, and batch GD yield identical solutions (0.4216 ± 0.0034), whereas SGD lags at 0.3853 due to stochastic noise under class skew. Logistic regression achieves 0.4300. Custom NumPy models surpass the linear ceiling: histogram GBM reaches CV F1 0.4354, and the 10-seed MLP reaches 0.4334. Their 50/50 blend achieves the best performance (CV F1 0.4368, held-out F1 0.4340), scoring 0.447 with 87.7% accuracy on AIcrowd.

Feature importance. Evaluating permutation drops in test AUC [10] reveals the primary predictive drivers: age (0.035 drop), self-reported general health (0.021), sex (0.018), high cholesterol (0.009), and prior stroke history (0.008). Direct risk lifts corroborate this: a prior stroke multiplies coronary disease risk by 4.24, and ≥ 5 chronic conditions carry a 3.72-fold risk lift.

Table II
CLEAN SCREENING MODEL VS. LEAKY COMPETITION VARIANT ACROSS 5-FOLD CV, HELD-OUT DATA, AND AICROWD.

Variant	CV F1	Held-out	AIcrowd	Acc.
Clean (Top 14%)	0.4368 ± 0.0026	0.4340	0.447	87.7%
Leaky (Top 13%)	0.4464 ± 0.0032	0.4445	0.457	88.3%
Gain (Δ)	$+0.0096$	$+0.0105$	$+0.0100$	$+0.6\%$

V. TARGET LEAKAGE ANALYSIS

Seven survey questions directly reference post-event care or secondary prevention: cardiac and stroke rehabilitation (`HAREHAB1`, `STREHAB1`), daily aspirin usage, and rationales (`CVDASPRN`, `RDUCHART`, `RDUCSTRK`, `RLIVPAIN`, `ASPUNSAF`). Notably, all 615 development respondents reporting cardiac rehabilitation (`HAREHAB1`) had positive heart diagnoses (and 158/158 on held-out data), representing severe target leakage [3].

As shown in Table II, incorporating these seven features boosts classification F1 by ≈ 0.010 across all models, elevating AIcrowd test F1 to 0.457 (88.3% accuracy). However, this reflects reverse causality. A clinical screening model serves undiagnosed individuals before disease onset; because undiagnosed individuals are never prescribed cardiac rehabilitation, these responses do not exist in genuine screening populations. We strictly exclude these leaky questions from our official model to preserve screening validity, while documenting the leaky model as a competition benchmark (`run.py -with_leakage`).

VI. DISCUSSION

All clean models plateau within 0.42–0.44 F1, reflecting an intrinsic information ceiling in self-reported telephone surveys. Error analysis reveals that false negatives are largely "silent" individuals: 56% are female, 40% lack diagnosed hypertension, 36% report excellent/good health, and 96% have no stroke history. On paper, these individuals exhibit zero classical warning signs, offering virtually no statistical signal in survey responses. Breaking this ceiling would require objective clinical measurements (e.g., lipid panels, ambulatory blood pressure, or ECG). Furthermore, public benchmark scores contain non-trivial variance: a minor 0.5% shift in test positive prevalence moves F1 by ≈ 0.01, showing that leaderboard differences partly reflect sampling noise.

VII. CONCLUSION

Our pure-NumPy pipeline, combining codebook cleaning, histogram GBMs, and an MLP, achieves 0.434 held-out F1 and 0.447 on AIcrowd (87.7% accuracy). We uncover severe target leakage in seven post-event questions that inflates F1 to 0.457 (88.3% accuracy), and exclude them from our primary model to ensure clinical validity for undiagnosed screening populations.

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